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Optimal modelling and artificial intelligent control of ultra-supercritical power unit: a survey

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ABSTRACT

Ultra-supercritical (USC) units, which represent the forefront of clean coal technology, offer superior efficiency and significantly lower emissions compared to conventional units. However, their operation under extreme conditions introduces complex technical challenges that necessitate sophisticated solutions. The existing literature review indicates a notable lack of dedicated reviews on modelling and control strategies for USC units. To address this gap, the study begins by contextualizing the operational challenges, technological significance, and fundamental principles of USC units within contemporary energy systems. Subsequently, a comprehensive examination of modelling methodologies is provided, covering both first-principles and experimental approaches, along with a critical discussion of their applicability and limitations. The review further analyzes advanced control strategies, ranging from classical linear and nonlinear methods to cutting edge artificial intelligence based techniques, assessing their performance in enhancing operational efficiency and flexibility. Finally, promising future research directions are outlined, offering insights into next generation modelling and control paradigms that may further optimize USC unit performance. Unlike earlier surveys focussed on conventional coal-fired units, this work specifically addresses the unique requirements and emerging intelligent solutions tailored to ultra-supercritical technology.

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advanced control strategies;
future research directions

1. Introduction

Fossil fuels, including petroleum and natural gas, constitute non-renewable energy resources. While global reserves remain relatively abundant in the short term, these finite resources are being rapidly depleted through intensive exploitation. Unchecked consumption inevitably leads to resource exhaustion. More critically, fossil fuel combustion emits hazardous pollutants – notably sulfur dioxide SO₂, hydrogen sulfide H₂S, nitrogen oxides NO_x, and carbon dioxide CO₂ – with CO₂ emissions being the primary driver of anthropogenic greenhouse effects (Andres et al., 2013). To mitigate these climate risks, 178 signatory nations adopted the landmark Paris Agreement, establishing a unified framework for climate action beyond 2020. The agreement's central objective aims to limit global temperature rise to well below 2 °C above pre-industrial levels, with aspirational targets of 1.5 °C (Paris Agreement, 2015).

Notwithstanding remarkable renewable energy advancements, thermal power generation (predominantly coal and gas-fired) maintains its pivotal role in global electricity systems. In 2024, China's thermal power infrastructure reached 1.44 billion kilowatts in installed capacity (43.1% of national total), generating 6.2 trillion kWh (62.6% of total output), demonstrating its critical function in grid stabilization and renewable energy intermittency mitigation (China Electricity Council, 2024). Across emerging economies like India and Southeast Asia, thermal power dominates the energy mixes, exceeding a 50% share. China has pioneered low-carbon transformation through comprehensive ultra-low emission retrofits, covering more than 95% of coal-fired units, and enhanced operational flexibility, with 50% of units achieving deep load cycling, thereby balancing energy security with carbon neutrality commitments (Ministry of Ecology and Environment, 2024). Looking forward, thermal power will evolve into a strategic balancing resource, supporting global energy transition during

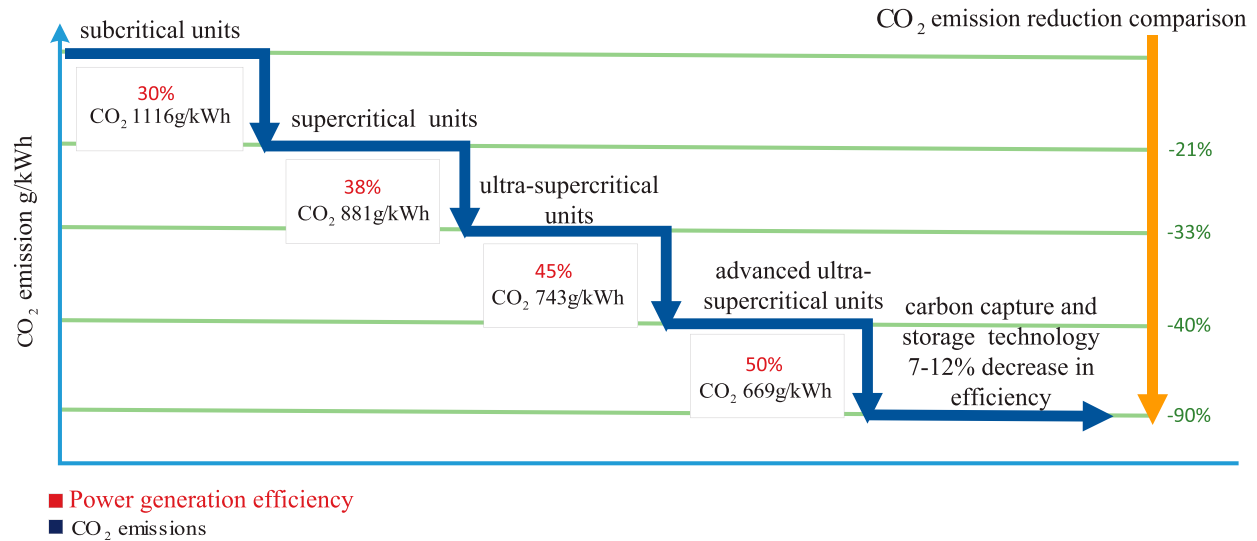


Figure 1. Comparison chart of CO₂ emission and efficiency of coal-fired power plants (X. L. Zhang et al., 2014).

this critical interim period. Strategic planning and implementation of more robust policies and measures to advance global emission peaking ahead of schedule, in response to climate change imperatives and sustainable development requirements, represents a critical challenge that must be urgently addressed in seeking of green, low-carbon development (Q. Zhang et al., 2022).

The concept of ultra-supercritical conditions originates from the thermodynamic critical point of water, which is 22.115 MPa pressure and 374.15 °C temperature. Thermodynamically, the supercritical state represents a phase where the distinction between liquid-vapor phases disappears, while ultra-supercritical conditions denote an even higher operational regime beyond conventional supercritical parameters. USC units typically operate at steam pressures exceeding 25 MPa and temperatures surpassing 580 °C, leveraging the direct correlation between steam cycle efficiency and operational parameters to deliver superior performance characteristics compared to conventional supercritical and subcritical units (X. Li et al., 2022). This advanced technology offers multiple competitive advantages: enhanced generation efficiency through optimized thermodynamic cycles, increased unit capacity enabled by higher energy density operation, improved operational reliability via advanced materials and control systems, and significant energy conservation coupled with environmental protection benefits through reduced fuel consumption and lower emissions per unit output. The combination of these features positions ultra-supercritical technology as the state-of-the-art solution in modern thermal power generation, achieving unprecedented levels of performance while addressing critical environmental concerns in the energy sector.

Figure 1 illustrates the comparative parameters of ultra-supercritical, supercritical, and subcritical power units. The most advanced ultra-supercritical units achieve maximum thermal efficiency of 50%, representing a 20% improvement over subcritical units and 12% higher than supercritical units. Furthermore, advanced ultra-supercritical units demonstrate coal consumption of 669 kg/MWh, reducing carbon emissions by 40% compared to subcritical units and 24% lower than supercritical units. While implementing carbon capture and storage technology may decrease unit efficiency by 7–12 percentage points, it enables up to 90% emission reduction compared to subcritical units. Consequently, the vigorous development of ultra-supercritical power generation technology has emerged as an effective approach to enhance economic efficiency while reducing pollutant emissions.

While USC units deliver superior efficiency, low emissions, and large capacity advantages, they present significant control challenges due to inherent nonlinear dynamics and strong multivariable coupling under high operating parameters. The system exhibits complex interactions where minor perturbations in feedwater flow, fuel input, or turbine valve position can substantially impact critical steam parameters and power output. This interdependence creates challenging dynamic behaviour, as turbine valve adjustments influence both feedwater flow and main steam pressure, while fuel modifications simultaneously affect thermal efficiency and emission profiles (L. Pan et al., 2020; Ping et al., 2024).

This operational complexity is further amplified when USC units support renewable energy integration through peak-shaving and frequency regulation. Such functions require operation across extended load ranges, intensifying system demands. During such flexible operation, the once-through boiler demonstrates strong nonlinear characteristics, with thermodynamic properties varying significantly across different operating modes. Technical challenges are particularly acute during transitional operations, where the combined effects of parameter coupling and nonlinear dynamics necessitate advanced control architectures to simultaneously maintain stable steam temperature/pressure, optimal combustion efficiency, emission compliance, and equipment safety margins. Consequently, the development of sophisticated multivariable control strategies represents a critical requirement for maximizing the operational potential of USC technology in modern power systems undergoing energy transition.

Given the increasing operational demands placed on USC units in modern energy systems, this paper presents a critical and systematic review of existing literature to establish a comprehensive understanding of current challenges and opportunities. Unlike prior surveys focussed on conventional coal-fired units, this work specifically addresses the unique requirements and emerging intelligent solutions tailored to USC technology. It systematically examines the distinct operational challenges of USC units, such as intensified nonlinearity and strong multivariable coupling under once-through boiler operation, with a focus on the advanced modelling and control paradigms designed to overcome them, including high-fidelity first-principles models, hybrid neural networks, and AI-based control strategies. By synthesizing current knowledge and mapping future research directions, this paper serves as both a comprehensive reference on existing USC control technologies and a strategic roadmap for developing next-generation control systems in thermal power generation.

To lay the groundwork for discussing these modelling and control challenges, it is essential to first elucidate the fundamental power generation process and the inherent dynamic characteristics of USC units, which is the focus of the following section.

2. Coal-fired ultra-supercritical power generation process

USC units are typically designed with single-unit capacities ranging from 600 MW to 1350 MW, with the 660 MW and 1000 MW configurations being the most prevalent in commercial power generation. These units operate under extreme steam conditions, reaching pressures as high as 30 MPa and temperatures up to 600 °C. By combining these high operating parameters with large capacity designs, thermal efficiency can reach up to 46%, marking a substantial improvement in fossil fuel power generation technology.

The standard engineering layout for the USC unit is shown in Figure 2. The USC power generation process represents the pinnacle of thermal power technology, achieving remarkable efficiency through optimized thermodynamic cycles. In the boiler-turbine system, raw coal is first pulverized into fine powder and injected into the furnace for combustion. Simultaneously, feedwater is preheated and circulated through waterwall where it absorbs intense heat, transforming into saturated steam. This steam then passes through superheaters to reach temperatures exceeding 600 °C before entering the multi-stage turbine system. The turbine arrangement, comprising high-pressure, intermediate-pressure, and low-pressure sections, efficiently converts the steam's thermal energy into mechanical rotation that drives the generator.

Demonstrated thermodynamic efficiency and emission performance advantages over conventional supercritical and subcritical units come with substantially greater operational complexity in USC power plants, attributable to extreme thermodynamic conditions and large-scale deployment. The enhanced performance of USC technology introduces greater sophistication in the following aspects (Esmaeili & Moradi, 2023):

- *Strong coupling:* The thermodynamic behaviour of USC units differs fundamentally from conventional drum-type boilers due to operation beyond the critical point (22.115 MPa, 374.15 °C). In USC units, the disappearance of the liquid-vapor phase boundaries creates an inherently coupled process where steam generation progresses through three characteristic phases: the heat absorption and evaporation phase, the steam absorption of latent heat phase, and the superheater heat absorption phase. This physical transformation establishes tight interdependencies among fuel flow dynamics, feedwater regulation, and turbine valve operation, resulting in strong nonlinear coupling effects that manifest as complex multivariable control challenges and significant operational constraints. The absence of conventional phase boundaries fundamentally alters the plant's dynamic characteristics, particularly during load transitions where coupling

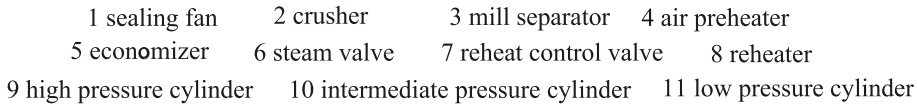


Figure 2. Power generation process of ultra-supercritical units (C. Cheng et al., 2024).

effects become most pronounced, necessitating advanced control strategies to ensure stable and efficient operation across the entire operating range.

- **Strong nonlinearity:** The operational dynamics of USC boilers exhibit inherent and pronounced nonlinearity due to the dramatic variations in thermophysical properties of the working fluid under different operating conditions. During load cycling operations across the wide pressure range of 10–25 MPa, the medium undergoes significant and rapid changes in key thermodynamic properties including specific heat capacity, specific volume, and enthalpy. These variations are not linear but rather demonstrate complex, coupled relationships with temperature and pressure parameters. The heat transfer characteristics and flow dynamics of the working fluid are particularly sensitive to these thermodynamic property changes, resulting in strongly nonlinear correlations between temperature gradients, pressure differentials, and heat flux distributions. Furthermore, the flow resistance characteristics and phase transition behaviours exhibit substantial nonlinear dependencies on the operating pressure and mass flow rate. These fundamental thermodynamic variations create substantial nonlinear multivariable interdependencies. This nonlinear behaviour significantly increases the challenges in both system modelling and control strategy development for USC units, requiring advanced mathematical approaches to accurately capture the process dynamics and maintain stable operation across the entire load range.

The analysis demonstrates that USC units operate under exceptionally demanding conditions, particularly when addressing the challenges of large-scale renewable energy integration and modern grid dispatch requirements. Current control systems remain inadequate to meet these operational demands, highlighting the critical need for advanced modelling and control technologies.

The intricate nonlinear dynamics and strong coupling effects outlined above necessitate sophisticated modelling techniques to accurately characterize system behaviour. Therefore, the next section provides a comprehensive review of the modelling methodologies employed for dynamic performance characterization of USC units.

3. Modelling methodologies for ultra-supercritical units in dynamic performance characterization

USC units represents a paramount challenge in contemporary energy engineering, characterized by intricate nonlinear dynamics, complex multi-variable, and multi-loop coupling operating across disparate timescales.

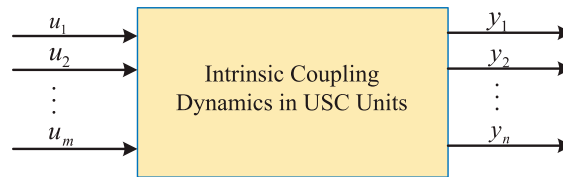


Figure 3. Input and output of ultra-supercritical units.

The mathematical modelling serves as the foundational pillar for cutting-edge research in operational optimization, advanced control architectures, comprehensive performance benchmarking, intelligent fault diagnostics, and high-fidelity virtual simulation platforms. As a quintessential interdisciplinary endeavor at the nexus of thermal science, control engineering, and computational mathematics, breakthroughs in this domain hold transformative potential for next-generation intelligent power plants and sustainable energy systems.

The thermodynamic system modelling process for thermal power units essentially involves the mathematical derivation of a multi-input/multi-output (MIMO) system representation $f(\cdot)$, as illustrated in Figure 3. The development of mathematical models for USC units primarily involves two fundamental approaches: first-principles modelling and experimental modelling (Chandrasekharan et al., 2013; P. Chen et al., 2022; Kocaarslan & Cam, 2007a).

- *First-principles modelling:* This method formulates mathematical representations based on established scientific principles, thermodynamic laws, and conservation equations, systematically characterizing the physical processes within each subsystem.
- *Experimental modelling:* This data-driven approach derives models through empirical testing on operational systems. Depending on prior knowledge, it can be categorized into:
 - Complete Identification: Requires assumptions about system behaviour when no prior knowledge is available.
 - Partial Identification: Incorporates known system characteristics to guide model development.

3.1. First-principles modelling methodologies for ultra-supercritical units

First-Principles modelling constitutes a rigorous methodology grounded in fundamental thermodynamic principles, conservation laws, and heat transfer mechanisms, thereby developing mathematical representations of complex thermal-hydraulic processes. This approach requires comprehensive characterization of physical phenomena to establish high-fidelity functional models that accurately capture:

- Dynamic system behaviour across multiple timescales
- Nonlinear interactions between subsystems
- Disturbance propagation characteristics

In USC once-through boilers, both the waterwall and superheater are composed of numerous parallel tubes of equal length, functioning as surface heat exchangers. As the working fluid flows through these tubes, it continuously absorbs heat, thereby undergoing changes in its thermodynamic state.

Early first-principles studies were conducted by Laubli et al., who performed mathematical investigations on supercritical units through empirical experiments and time-based simulations (Laubli & Fenton, 1971). Their work conclusively demonstrated the rapid load-following capabilities of supercritical units, thereby validating the classical coordinated control performance underpinning the flexible operation. While the detailed derivations of boiler modelling were not explicitly elaborated, a lumped-parameter computational model was provided. Notably, the turbine, generator, fan, and other rotating components were consistently treated as a single flywheel, which stores rotational energy as speed increases. The simulation results robustly affirm the feasibility of supercritical unit as both a flexible baseload generation technology and a highly responsive solution for fast demand-tracking power generation.

Work with formula expression for power plant was pioneered by researchers including Åström, Bell, and Eklund in the 1970s (Åström & Eklund, 1972; Bell & Åström, 1979). Their work derived mathematical models of

varying complexity by applying fundamental conservation laws with different simplification approaches. Studies demonstrated that higher model accuracy typically requires more complex structures, involving additional unknown parameters and computational effort. A notable example is the 160 MW oil-fired drum boiler nonlinear model developed by Bell and Åström in 1987, as shown in Equation (1), which adopted a lumped-parameter approach and became a classic reference in the field (Åström & Bell, 2000; Bell & Åström, 1987). These foundational studies established important principles for balancing model fidelity with practical usability in industrial applications.

$$\left\{ \begin{array}{l} \dot{x}_1 = -0.0018u_2x_1^{9/8} + 0.9u_1 - 0.15u_3 \\ \dot{x}_2 = (0.073u_2 - 0.016)x_1^{9/8} - 0.1x_2 \\ \dot{x}_3 = (141u_3 - (1.1u_2 - 0.19)x_1)/85 \\ y_1 = x_1 \\ y_2 = x_2 \\ y_3 = 0.05(0.13073x_3 + 100\alpha_{cs} + q_e/9 - 67.975) \\ \alpha_{cs} = \frac{(1 - 0.001538x_3)(0.8x_1 - 25.6)}{x_3(1.0394 - 0.0012304x_1)} \\ q_e = (0.845u_2 - 0.147)x_1 + 45.59u_1 - 2.514u_3 - 2.096 \end{array} \right. \quad (1)$$

where u_1 represents the fuel flow rate (t/h), u_2 denotes the steam flow rate (t/h), u_3 is the feedwater flow rate (t/h), x_1 and y_1 indicate the drum pressure (kg/cm²), x_2 and y_2 represent the unit load (MW), x_3 stands for the steam-water density (kg/m³), y_3 shows the drum water level (m), α_{cs} is the steam quality, and q_e signifies the evaporation rate (kg/s). This model structure has been widely applied in coordinated control system design for drum boiler units (Feng et al., 2023; He & Lima, 2019; S. Liu et al., 2022). However, the model exhibits the following limitations:

- (1) Applicability constraints: Developed based on oil-fired units, incompatible with current units;
- (2) Control variable mismatch: Output variables (drum pressure/liquid density) demonstrate generational gaps with modern coordinated control systems (focussed on main steam pressure and load);
- (3) Technological generation disparity: Fundamental differences exist between the ultra-high pressure unit characteristics and current USC units.

In the 1990s, Mello's research on simplified modelling of coal-fired drum boilers incorporated the heat storage characteristics of the drum, water walls, and superheater, as well as the influence of differential pressure, resulting in improved model accuracy (Demello, 1991; Forli et al., 1991). Building upon this model, Zeng and Tian established an enhanced model for a subcritical coal-fired drum boiler through mechanistic analysis and experimental testing, with the following structure (L. Tian et al., 2004; Zeng & Liu, 2000; Zeng & Zhao, 2003):

$$\left\{ \begin{array}{l} \dot{x}_1 = -\frac{x_1}{120} + \frac{e^{-18s}\mu_1}{120} \\ \dot{x}_2 = (-0.25(x_2 - 0.005467x_1^{1.3})\mu_2 + 6.77x_1)/3266 \\ \dot{x}_3 = -(-x_3 + 0.25(x_2 - 0.005467x_1^{1.3})\mu_2)/12 \\ y_1 = x_2 - 0.005467x_1^{1.3} \\ y_2 = x_3 \end{array} \right. \quad (2)$$

where u_1 represents the fuel quantity (t/h), u_2 denotes the main steam valve opening (%), u_3 is the pulverized coal flow rate (t/h), x_1 indicates the drum pressure (MPa), x_2 and y_1 represent the unit load (MW), and x_3 signifies the main steam pressure (MPa).

Equation (2) represents the model of traditional steam drum boilers, current modelling methodologies for USC units exhibit significantly lower precision compared to conventional thermal power plants, primarily due to unresolved complexities in phase transition localization within boiler steam systems. The seminal

work adopted a fixed fuel-water ratio assumption, reducing the once-through boiler to an oversimplified dual-input-dual-output model that fundamentally neglected phase transition dynamics in the steam-water flow path (Inoue et al., 2000). This critical simplification resulted in substantially compromised model fidelity. More importantly, while modern coordinated control systems for once-through boilers require precise regulation of three key parameters – unit load (MW), main steam pressure (MPa), and intermediate point temperature (°C) – these rudimentary models only incorporate two outputs, rendering them fundamentally inadequate for advanced control system synthesis and optimization. It should be emphasized that the main steam temperature is the highest in the entire unit and needs to be strictly controlled. However, due to the considerable distance between the boiler inlet and the superheater outlet, direct control becomes challenging. Therefore, by utilizing the intermediate point temperature as a pre-control variable, combined with the staged temperature reduction in the superheater, effective control of the superheated steam temperature can be achieved. Shinohara et al. proposed a simplified state-space model for a third-order supercritical direct current boiler, which also encounters similar issues (Shinohara & Koditschek, 1995). The expression of the model is as follows:

$$\begin{cases} \dot{x}_1 = \frac{1}{V_f} u_1 - \frac{k_v p_s}{V_f \sqrt{x_3}} u_2 \\ \dot{x}_2 = \frac{h_{ec} - h_{sc}[p_f, T_f] - \tilde{h}}{V_f \eta_f} u_1 + \frac{k_v p_s \tilde{h}}{V_f \eta_f \sqrt{x_3}} u_2 + \frac{k_{fm} \Delta q_c}{V_f \eta_f} u_3 \\ \dot{x}_3 = \frac{k_v p_s (h_f - h_s)}{V_s \eta_s \sqrt{x_3}} u_2 + \frac{k_{sh} \Delta q_c}{V_s \eta_s} u_3 \\ y = \frac{k_v p_s (h_s - h_{cd})}{k_{ib} \sqrt{x_3}} u_2 \end{cases} \quad (3)$$

where

$$\begin{aligned} \eta &= \left(\rho \frac{\partial h_{sc}}{\partial P} [P, T] - 1 \right) \frac{\partial F_H}{\partial T} [T, \rho] + \rho \frac{\partial h_{sc}}{\partial T} [P, T] \\ \tilde{h} &= \left(\rho_f \frac{\partial h_{sc}}{\partial P} [P_f, T_f] - 1 \right) \frac{\partial F_H}{\partial \rho} [T_f, \rho_f] \\ h_{sc}[P, T] &= \beta_0 + \beta_1 P + \beta_2 T + \beta_3 P e^{-\alpha(T-T_0)P} + \beta_4 T e^{-\alpha(T-T_0)P} \\ F_H[T, \rho] &= \rho T (\alpha_0 + \alpha_1 \rho + \alpha_2 T + \alpha_3 \rho T + \alpha_4 T^2) \end{aligned} \quad (4)$$

herein, u_1 represents the feedwater flow rate (lb/h), u_2 denotes the main steam valve opening (%), $0 \leq u_2 \leq 1$, u_3 is the coal feed rate (lb/h), x_1 indicates the steam-water density in waterwall (lb/ft³), x_2 shows the steam temperature in waterwall (°C), x_3 signifies the steam temperature in superheater (°C), y represents the unit load (MW), h_{ec} is the feedwater enthalpy (Btu/lb), h_{cd} denotes the turbine exhaust enthalpy (Btu/lb), Δq_c is the coal heating value (Btu/lb), while V , P , T , and ρ stand for volume, pressure, temperature, and density respectively, with subscripts f for waterwall and s for superheater. k_v , k_{fm} , k_{sh} , k_{tb} representing volumetric, furnace normalization, superheater heat transfer, and turbine bypass coefficients respectively.

These models have been validated under single operating conditions, but their applicability across a wide load range remains unverified. Nevertheless, these work establishes a fundamental basis and provides guidance for future mechanistic modelling, with most subsequent literature building upon these foundational models. Gao et al. linearized the boiler-turbine system to enable rapid assessment of grid-level dynamics. The model demonstrated acceptable accuracy in both frequency-domain analysis and load disturbance tests when benchmarked against nonlinear references (Gao & Dai, 2011). However, this study relied entirely on linear assumptions and omitted calibration with actual operational data, resulting in significantly amplified errors under off-design conditions. Sreepradha et al. developed a low-order mechanistic model for a 210 MW unit by rigorously applying mass, energy, and momentum conservation principles alongside the Clausius-Clapeyron relation (Sreepradha et al., 2017). The model was validated using real-time field data for critical variables such as liquid level, temperature, and pressure. The results confirmed strong agreement between simulated and measured values, thereby addressing the lack of empirical support inherent in linearized models.

Although Sreepradha's method was calibrated with real-world data, it still lacks exploration of generalization capability. In contrast, Oko et al. established a rigorous dynamic modelling framework for subcritical power plants, demonstrating comprehensive validation against operational data coupled with performance analysis under diverse operating conditions (Oko & Wang, 2014). The developed model successfully captures the intricate coupling between combustion dynamics, steam generation processes, and power conversion characteristics, with particular emphasis on the critical fuel-air combustion and water-steam cycle interactions. Then this method is further extended to a 1000 MW USC unit based on physical laws and simplified assumptions, which demonstrated reasonable consistency for both static and dynamic simulations (J.-Z. Liu et al., 2015). The static calculations effectively covered a broader operational range from 505 MW to 1000 MW.

While hybrid modelling offers broad operational coverage, it remains constrained by the assumption of fixed unmeasurable parameters. Chandrasekharan et al. addressed this limitation by developing an integrated model for a 210 MW coal-fired boiler, employing an energy balance and thermodynamic principles-based model to analyze internal processes and examine the effects of varying feedwater temperature (Chandrasekharan et al., 2019). Wang et al. advanced this paradigm in a 660 MW USC unit by constructing a high-fidelity model on the GSE platform. They introduced a novel 'steady-state coal consumption + transient coal consumption increment' framework to inversely calibrate dynamic parameters using field data (C. Wang et al., 2017). The study revealed thermal inertia and control strategies as pivotal factors in coal consumption variability, further underscoring the inadequacy of fixed parameters in capturing real-world operation. Then this work on the same unit is extended by investigating the impact of intermediate point temperature setpoint variations on model accuracy (Y. Wang et al., 2019). By refining heat transfer coefficients through energy balance and thermodynamic equations, the enhanced model successfully replicated internal processes across diverse setpoints, thereby resolving the issue of parameter drift under varying operational conditions.

Conventional identification methods are primarily designed for standalone coal-fired units and become inadequate when applied to integrated systems involving gas turbines or other external components due to the substantial increase in parameters requiring identification. Zhou et al. tackled this challenge by developing a comprehensive multi-physics dynamic model that couples a 1000 MW USC unit with a gas turbine system, incorporating combustion dynamics, heat transfer processes, and control mechanisms (J. Zhou et al., 2024). They introduced an innovative adaptive parameter identification framework based on Gibbs free energy minimization principles and pressure-driven governing equations. Simulation results demonstrated this framework's exceptional capability to accurately simulate transient load responses across various operational modes, effectively overcoming the applicability limitations of traditional single-unit identification methods in complex multi-energy coupled systems.

Guo et al. established a pioneering coupled dynamic model integrating a 600 MW USC unit with an amine-based carbon capture system, introducing for the first time a dynamic parameterization methodology encompassing the complete absorption-desorption process (Guo, 2015). The model was rigorously validated across a 40%–100% load range, successfully capturing the dynamic interactions between steam extraction, reboiler thermal load, and power output. While this work successfully established the critical dependence of carbon capture system evaluation reliability on unit-side model accuracy – highlighting the capture system's sensitivity to power plant model fidelity – it did not investigate the host unit's operational characteristics during extreme transient scenarios. A dynamic model addresses startup challenges in 600 MW USC boilers, where operational deviations and transients cause significant emissions (Deng et al., 2017). Using a six-equation two-phase flow solution, it simulates evaporator dynamics from ignition to 30% load, demonstrating superior practicality versus conventional methods. As shown in Table 1, steady-state validation at 40% turbine heat acceptance showed excellent agreement with design specifications.

First-principles modelling demonstrates superior fidelity in USC unit simulations through thermodynamic laws and transport mechanisms, effectively capturing multi-scale dynamics and nonlinear phenomena, though challenged by parametric uncertainties and generational gaps in legacy systems. Empirical models achieve localized accuracy through data calibration but suffer error accumulation during off-design operations, while hybrid approaches balance physical principles with data assimilation yet require sophisticated calibration techniques. These complementary methodologies each present unique advantages in accuracy, applicability and adaptability, with optimal selection dependent on specific operational requirements. Table 2 summarizes the survey for coverage from 1971 to 2025.

Table 1. The comparison under 40% turbine heat acceptance condition.

Parameter	Simulation result	Design value	Unit	Relative error (%)
Furnace exit gas flow	346.95	346.95	kg/s	0.00
Economizer gas flow	211.11	211.11	kg/s	0.00
Feed-water flow	188.55	188.55	kg/s	0.00
Water wall entrance temperature	284	284	°C	0.00
Water wall entrance pressure	12.4	12.32	MPa	0.65
Water wall exit temperature	347	347	°C	0.00
Water wall exit pressure	11.93	12.08	MPa	1.24
Gas flow of platen superheater	346.95	346.94	kg/s	0.00
Platen superheater entrance temperature	424.8	424.97	°C	0.04
Superheater exit temperature	572	572	°C	0.00

Table 2. First-principles modelling review summary (coverage 1971–2025).

Principle: Mathematical representation for complex industrial processes based on fundamental thermodynamic principles, conservation laws, and heat transfer mechanisms

modelling methodology	Advantages	Disadvantages
Early modelling for thermal power plants Equation (1–4) (Laubli & Fenton, 1971–Inoue et al., 2000)	The accuracy of modelling for USC units is significantly lower than that of traditional thermal power plants	The accuracy of modelling for USC units is significantly lower than that of traditional thermal power plants
Modelling within a wide load range (Gao & Dai, 2011–Y. Wang et al., 2019)	Solved the problem of insufficient empirical support in linearized models and covered a wider operating range	Linear models key limitation: inadequate handling of multivariate system complexity in modern industry
Modelling of external components (Guo, 2015–J. Zhou et al., 2024)	Centering on the key physical processes of specific system, model fidelity is anchored to actual response	Ditto

3.2. Experimental modelling for ultra-supercritical power plants

The intricacies of multivariable systems in modern industrial settings often present significant challenges, making it challenging to establish accurate mathematical models through mechanistic analysis alone. Overly simplified modelling assumptions may further compromise model precision, necessitating the adoption of advanced system identification techniques for effective model development. Data driven system identification has gained widespread adoption due to its diverse methodologies, which can be applied across various scenarios. These methodologies are equally effective when a mathematical description of the system dynamics is available (white-box modelling) or when only input-output data are accessible without prior knowledge of the underlying physics and dynamic behaviour (black-box modelling). Moreover, data-driven modelling framework facilitates gray-box modelling, where parameters are estimated from data within a predefined model structure.

3.2.1. Linear system identification approaches

Traditional system identification techniques, such as the least squares method, gradient descent method, and maximum likelihood estimation, are effective and often the preferred approach for capturing small-signal responses, such as those observed in combined-cycle gas turbine power plants (Mohamed & ZaTer, 2020; Rajak & Pudur, 2025; Yan et al., 2015). However, modelling USC units presents significantly greater complexity due to their inherently slower dynamics under supercritical conditions, where the thermodynamic properties of steam and water exhibit non-linear variations, making it difficult to accurately predict system behaviour over time. To implement advanced controllers, such as advanced digital control and predictive control, aimed at enhancing control system performance, a statistical model that accurately characterizes the dynamic processes is essential (McMasters, 2002; H. Tian & Wang, 2012; Toyoda & Wada, 2010).

As mentioned earlier, modern USC units exhibit complex characteristics such as multivariate coupling and strong nonlinearity, rendering conventional identification methods inadequate for accurately representing their intricate operational dynamics using a single model. To address this challenge, advanced intelligent modelling approaches have been introduced. One of the pioneering efforts dates back to the dynamic local linear model network proposed by Prasad et al., which employs a partitioned operating regime strategy – dividing the entire operational domain into multiple sub-regions for systematic identification (Prasad et al., 1998). This

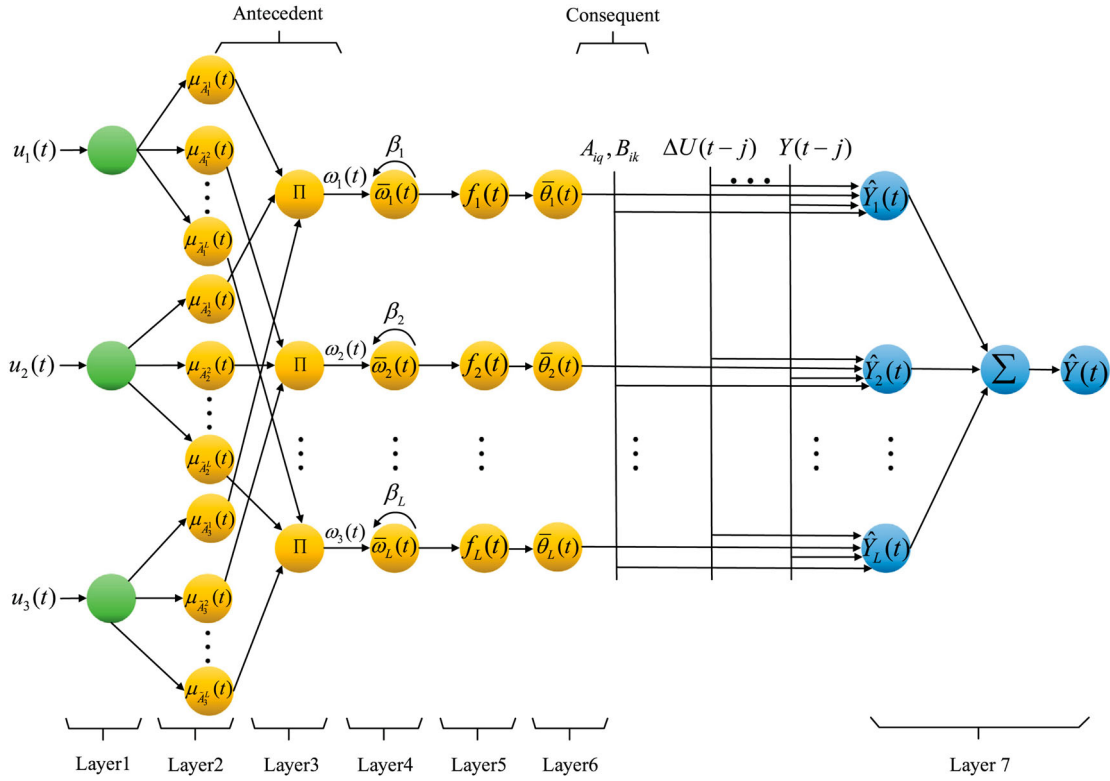


Figure 4. Structure of the FNN (M. Zhao et al., 2023).

method significantly enhances long-term prediction accuracy for key parameters, such as main steam temperature and pressure, which are crucial for maintaining operational stability and optimizing the performance of USC units under varying load conditions.

Based on the partitioning-based methodology, numerous researchers have advanced this modelling approaches for USC units. Liu et al. conducted dynamic experiments on boiler systems and established a linguistic representation for the main steam temperature model (X. J. Liu et al., 1999). A Gaussian equidistant midpoint partitioning system was employed to model steam boiler dynamics, with fuzzy logic utilized for the final model representation. Subsequently, fuzzy neural networks (FNNs) were introduced to independently model the main steam temperature and the multivariate output system of USC units, demonstrating superior predictive performance (X. Liu et al., 2009; X. Liu & Kong, 2013; X. J. Liu & Chan, 2006). The structure of FNN with inputs $U(t) = [u_1(t), u_2(t), u_3(t)]^T$ and outputs $Y(t) = [y_1(t), y_2(t), y_3(t)]^T$ is illustrated in Figure 4. Specifically, the input variables u_1 , u_2 , and u_3 are defined as fuel flow, governor valve opening, and feedwater flow, respectively. The output layer variables Y correspond to electric power, main steam pressure, and steam temperature. The functions of each layer are: the input layer (Layer 1) receives the system variables, the fuzzification layer (Layer 2) employs Gaussian membership functions for fuzzy mapping, the rule layer performs fuzzy inference (Layers 3–5), the normalization layer (Layer 6) computes normalized firing strengths, and the output layer (Layer 7) produces the final weighted inference result. This approach has been subsequently adopted by multiple research groups for USC unit modelling, with consistently favorable outcomes. A critical observation is that while the global architecture employed FNNs, local operational regimes maintained their representation through linear controlled autoregressive moving average (CARMA) formulations (Alitasb & Salau, 2024; Kong et al., 2015, 2016; Moon & Lee, 2011; M. Zhao et al., 2023).

In addition to the CARMA models, many researchers have introduced linear state-space model to simulate the dynamic process of USC units. For instance, a comparative study between subspace identification and the prediction error method was conducted, where a gas turbine power plant was modelled as a linear system (Mohamed et al., 2015). Good agreement with dynamic transient trends was observed. Then, a linearized time-varying model based on the lumped parameter approach was proposed, in which direct energy balance coordinated control was implemented to enhance the system's load-following capability (L. Sun et al., 2016).

Table 3. Experimental modelling review summary (linear model, coverage 1998–2025).

Principle: Treating the object as a 'black box', the non-mechanistic mathematical model is established by designing input-output experiments and employing system identification algorithms

Identification approaches	Advantages	Disadvantages
Traditional system identification techniques (Yan et al., 2015–Rajak & Pudur, 2025)	Computationally simple, suitable for linear systems, and low-cost for engineering implementation	Only capable of linear approximation, unable to capture high-order couplings and dynamic switching
Partitioned operating regime strategy (CARMA model) (Prasad et al., 1998–Alitasb & Salau, 2024)	Decomposing nonlinear systems into multiple local linear subdomains, reducing model complexity while preserving interpretability	Lacking theoretical support, the smooth articulation and stability between sub-models still require additional compensation strategies
Partitioned operating regime strategy (state-space model) (Mohamed et al., 2015–Javed et al., 2021)	Ditto	Ditto
Adaptive modelling (Wu et al., 2018b–Kestering et al., 2023)	Real-time parameter updating enables rapid compensation for disturbances such as coal quality fluctuations and load step changes	When operating conditions expand substantially or equipment aging causes sharp non-linearity increases, the union of model fails to cover the full operating domain

Additionally, linear state-space models were developed based on input-output data from USC units to predict future CO₂ capture levels and reboiler temperature (X. Wu, Shen, Li, Wang, Lawal & Lee, 2018a; X. Wu et al., 2019). These parameters were expressed as functions of lean solvent flow, reboiler steam, and flue gas flow rates. The models were identified for operating points characterized by a reboiler temperature of 386 K and a CO₂ capture rate of 70%. Javed et al. presented a subspace identification-based linear modelling framework for H_∞ multivariable controller synthesis in underground coal gasification plants (Javed et al., 2021). The proposed method successfully captures the nonlinear intricacies of subsurface gasification processes, yielding measurable improvements in plant-wide operational performance

While linear modelling remains a computationally efficient and reliable approach for conventional thermal plants, the inherent nonlinearity and process complexity of USC units necessitate adaptive modelling frameworks capable of online modification for real-time control applications. Research (X. Wu, Shen, Li, Wang & Lawal, 2018b) demonstrates this through subspace-derived multiple MPC implementation, where flue gas disturbance analysis enables more flexible USC operation. Parallel work advances MMPC design for coal pulverizing systems, combining first-principles dynamics with state-space multi-model inference to achieve offset-free performance (Arroyo et al., 2022; Kestering et al., 2023; Liang et al., 2018).

While the above linear/adaptive strategies offer valuable insights, they inevitably encounter accuracy ceilings when the operating domain expands or the process exhibits strong nonlinearity. Table 3 summarizes the survey for coverage from 1998 to 2025. To transcend these limitations, researchers have turned to artificial neural networks that intrinsically possess universal approximation capabilities. The following subsection reviews the research of nonlinear neural-network-based modelling approaches.

3.2.2. Nonlinear neural network models

Artificial neural network (ANN), with the inherent capability to represent arbitrary nonlinear mappings, have emerged as a transformative approach for nonlinear system modelling (Mohan George et al., 2024). The integration of ANN models into nonlinear system identification leverages the universal function approximation properties to accurately capture input-output relationships of practical systems, while the self-learning and adaptive characteristics enable the development of computationally tractable learning algorithms suitable for engineering implementation. In recent years, the application of neural networks to the identification of strongly nonlinear systems such as USC units has become an active research frontier in the ANN community. Current approaches can be categorized into three paradigms: (1) purely data-driven neural network modelling, (2) hybrid mechanism-neural modelling, and (3) adaptive online modelling frameworks (Benosman, 2018).

Data-driven neural networks (NN) have become a fundamental approach for modelling USC units, leveraging their model-free nature and strong nonlinear fitting capabilities. This method bypasses complex mechanistic modelling by learning input-output relationships directly from operational data, especially valuable for USC systems with intricate thermo-fluid dynamics.

Smrekar et al. established a high precision ANN-based model by developing and integrating separate boiler and turbine sub-models, enabling accurate prediction of power output with minimal input parameters

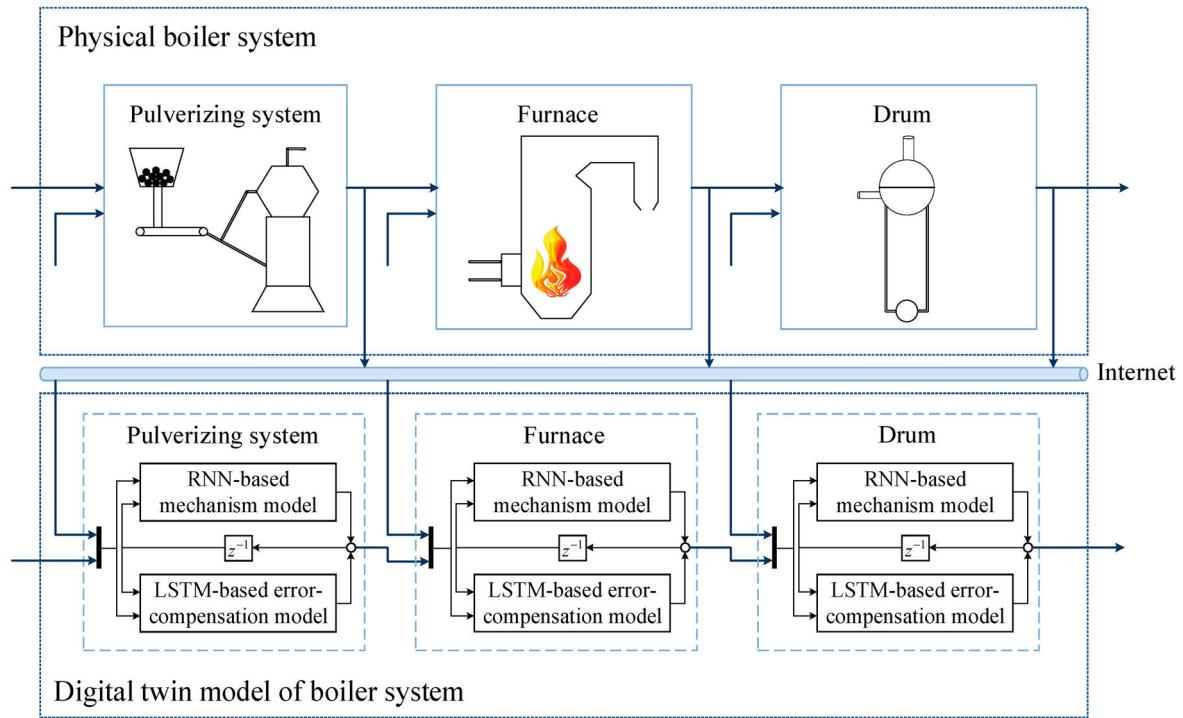


Figure 5. Process of hybrid neural modelling (Y. Zhao et al., 2023).

(Nistah et al., 2018). Compared to conventional models, this approach demonstrates superior responsiveness and enhanced adaptability for real-time updates. Then, a high-performance deep CNN prediction model is developed by ingeniously designing two lightweight modules incorporating 1D convolutional layers and channel-wise operations (N. Li & Hu, 2020). Utilizing operational data and coal quality characteristics from a USC boiler as inputs, the raw data were transformed into multivariate time series for modelling. The resulting model demonstrated exceptional predictive performance, achieving a remarkably low RMSE of 0.25 mg/Nm^3 on the test set. Zhou et al. proposed a monitoring approach for CO_2 emissions from coal-fired power units based on a heterogeneous neural network (S. Zhou et al., 2023). This method establishes a quantitative relationship between net power generation and emission intensity, enabling high-precision estimation solely through grid-side electricity meter data.

The above models focus on single-output formulations and largely overlook the inherent coupling among multiple operating parameters, thereby limiting their applicability to full-scale characterization of USC units. Significant advancements have been achieved through the development of a novel neural network framework specifically designed for multivariate output modelling in USC units (Y. Cheng et al., 2018; Cui et al., 2020; X. Liu et al., 2020; Sharma et al., 2023). Their innovative architecture incorporates adaptive feature extraction and cross-parameter correlation learning, enabling simultaneous high-accuracy prediction of multiple inter-dependent operational parameters. This pioneering approach demonstrates superior modelling fidelity and comprehensive system characterization.

Hybrid neural modelling combines physics and data science to effectively model USC complex dynamics. As shown in Figure 5, the input and output variables are the same as those in Figure 4. The hybrid model consists of two main components: an RNN-based mechanism model layer, which describes the dynamic behaviour of the boiler system and enables parameter updating, and an LSTM-based error compensation layer, which corrects the residual prediction errors of the mechanism model to improve overall accuracy. This approach integrates thermodynamic principles with neural networks, overcoming limitations of both pure data-driven and traditional mechanistic methods.

A hybrid data mechanism modelling approach is proposed for steam turbine control performance monitoring under renewable integration challenges (J. Yu et al., 2020). This approach enables: (1) flow rate estimation

with limited measurements and (2) full range characteristic calculation. Zhang et al. proposed an hybrid mechanism data modelling approach for gas turbines based on the dominant factor method (Y. Z. Zhang et al., 2021). By integrating thermodynamic principles with comprehensive analysis of operational data, they established a sophisticated mechanism data fusion model. This model demonstrates exceptional accuracy, with all calculated results exhibiting average relative errors below 0.8%. A novel RNN-LSTM hybrid modelling approach is developed for boiler system digital twins to overcome traditional mechanism model limitations (Y. Zhao et al., 2023). The method integrates RNN-based mechanistic representation for interpretability with LSTM networks to capture unmodeled dynamics, while establishing neuron state relationships for time-varying parameters. Jiang et al. developed a hybrid mechanism data modelling approach for complex solvent carbon capture systems, validated on a 1000 MW plant with 0.2–8% error in key parameter estimation (Jiang et al., 2025). Yang et al. proposed a stabilization method for control systems that is independent of the attack model. This method targets control systems that employ a jump-like time-ordering delivery scheduling protocol, aiming to enhance the system's stability in the face of unknown disturbances (H. Yang et al., 2023). Besides, a novel semantic-based multi-packet parallel transmission scheme is introduced, which can quickly process and transmit data, greatly improving the real-time performance of the system, which is particularly important for industrial control systems that require rapid response (H. Yang et al., 2025).

Adaptive online modelling has emerged as a cutting-edge approach to address the time-varying dynamics of USC units, achieving significant improvements in dynamic modelling accuracy across wide load ranges through real-time integration of operational data and model updating mechanisms. This paradigm overcomes the limitations of traditional static models by incorporating three key innovations: sliding time window strategies for dynamic parameter updates to accommodate coal quality variations and equipment aging, incremental learning algorithms for online model optimization, and digital twin integration for virtual-physical cross-validation to enable real-time state estimation for coordinated control systems.

Tang et al. proposed an innovative dynamic correction prediction model to address the challenges of time delay and strong nonlinearity in NO_x emission forecasting for USC units (Z. Tang et al., 2022). The method integrates maximum information coefficient and dataset reconstruction, combines lasso and relief algorithms for adaptive feature selection, and ultimately establishes an extreme learning machine model incorporating dynamic error correction. Then a dynamic optimization strategy based on neural networks combined with particle swarm optimization algorithm is employed in Blackburn et al. (2022). By establishing a dynamic physical boiler model incorporating multiple fuel/air injection points, the approach achieves up to 6.05% improvement in thermal efficiency compared to baseline operating conditions. Furthermore, an enhanced error-adaptive extended state kalman filter model predictive control methodology is introduced to address the challenges of strong coupling and inherent inertia in USC units (G. Chen et al., 2025). By integrating a real-time deviation-based dynamic error weighting mechanism with exponential smoothing into the BP neural network, the proposed approach demonstrates exceptional control performance and operational superiority. A novel deep neural network-based online reconstruction methodology is proposed to overcome the limitations of conventional image detection systems. By utilizing offline simulation to generate a comprehensive training dataset and establishing a complex DNN model, real-time temperature field reconstruction can be achieved (Xue et al., 2024).

Table 4 summarizes the survey coverage for nonlinear modelling from 2018 to 2025. In practical engineering applications, the selection of an appropriate modelling methodology for USC units should be guided by the specific objectives, data availability, and real-time operational requirements of the system.

First, mechanism-based (first-principles) models are most suitable for high-fidelity simulation, performance evaluation, and controller design, where physical interpretability and accuracy are of primary importance. Second, data-driven models, including neural networks and system identification techniques, are recommended for real-time prediction, operational optimization, and adaptive control, especially when sufficient operational data are available but detailed physical information is limited. Third, hybrid modelling approaches that integrate physical mechanisms with data-driven learning provide an effective compromise between accuracy and adaptability. Such hybrid models are particularly advantageous for wide-load, strongly coupled, and time-varying USC unit operations, enabling intelligent modelling and digital twin applications.

Overall, these three categories of modelling strategies complement each other, and the optimal choice should depend on the intended application scenario, whether for design, operation, or real-time optimization.

Table 4. Experimental modelling review summary (nonlinear model, coverage 2018–2025).

Principle: Through multi-level nonlinear transformations and distributed representations, the data is mapped into a high-dimensional feature space, approximating arbitrarily nonlinear relationships

Modelling methodology	Advantages	Disadvantages
Data-driven neural networks (Nistah et al., 2018–S. Zhou et al., 2023)	Model-free data-driven nature, powerful nonlinear fitting capability, efficient computing architecture	Focusing on single-output formulations while largely overlooking the inherent coupling among multiple operating parameters
Hybrid neural modelling method (Y. Cheng et al., 2018–Jiang et al., 2025)	Combining physics and data science to effectively model USC complex dynamics	Facing computing complexity in balancing fidelity and neural network adaptability, especially for parameter-coupled systems
Adaptive online modelling method (Z. Tang et al., 2022–Xue et al., 2024)	Enabling real-time tracking of USC unit dynamics through sliding window updates and incremental learning	Requiring high-frequency data acquisition infrastructure and facing stability-complexity tradeoffs, particularly during rapid load transitions

With the foundation of modelling methodologies established, the discussion now turns to the core objective: enabling effective control. The accurate models discussed herein serve as the cornerstone for developing the advanced control strategies that are essential for managing the complex dynamics of USC units. These strategies will be thoroughly introduced in the following section.

4. Control strategies of ultra-supercritical units

The fundamental imperative governing the operation of coal-fired power units resides in the precise delivery of demanded electrical power while maintaining stringent operational constraints across the entire energy conversion chain. This operational mandate necessitates the synergistic optimization of interconnected subsystems, including but not limited to pulverizing mills, steam generation systems, and turbogenerator assemblies, all of which must operate at peak thermodynamic efficiency while adhering to exacting safety regulations. Contemporary control architectures must be predicated upon high-fidelity models that capture the intricate dynamics governing plant operations. Only through such rigorous modelling frameworks can advanced control strategies achieve the dual objectives of maximizing power generation efficiency and minimizing specific fuel consumption while ensuring emissions compliance with increasingly stringent environmental regulations (Badgujar, 2016; Chandrasekharan et al., 2013).

While conventional PID control architectures have historically dominated industrial applications with implementation simplicity and computational parsimony, their limitations become critically apparent when confronted with the multifaceted challenges of modern power generation systems (Clarke, 1984; Wichert & Dhaliwal, 1978). Today's thermal plants must not only navigate complex operational regimes with tight coupling between process variables but also fulfill critical grid-support functions including dynamic load following, primary frequency response, and ultra-low emission maintenance, all while accommodating the stochastic variability introduced by renewable energy integration. These challenges are particularly acute in USC units, where the absence of phase separation boundaries in once-through boiler designs creates highly nonlinear, multivariable process dynamics. The strong cross-coupling between combustion parameters, feed-water chemistry, and steam conditions manifests as complex, state-dependent interactions that conventional single-input-single-output control paradigms cannot adequately address (Burgasi et al., 2021).

The pursuit of rapid and effective control methodologies for USC units has engendered significant research efforts from both industrial practitioners and academic investigators. As shown in Figure 6, building upon extensive theoretical analyses and field validations, contemporary control strategies have achieved remarkable advancements in operational performance. The following synthesis systematically categorizes and evaluates the state-of-the-art control paradigms that have demonstrated efficacy in modern coal-fired power generation systems.

4.1. Control methodologies based on linear system theory

Linear system theory forms the basis for the most mature and extensively applied methods in control engineering. Research in this area is relatively advanced and enjoys widespread application. Notable studies include PID control and parameter self-tuning, robust control and tuning, predictive control, among others.

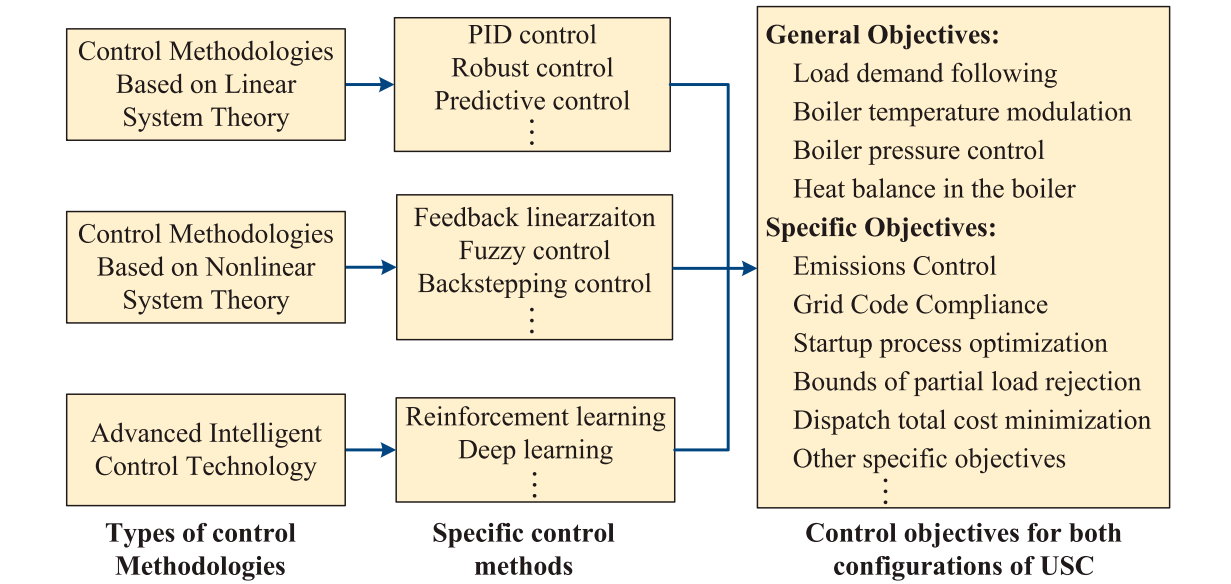


Figure 6. Different control strategies for USC units.

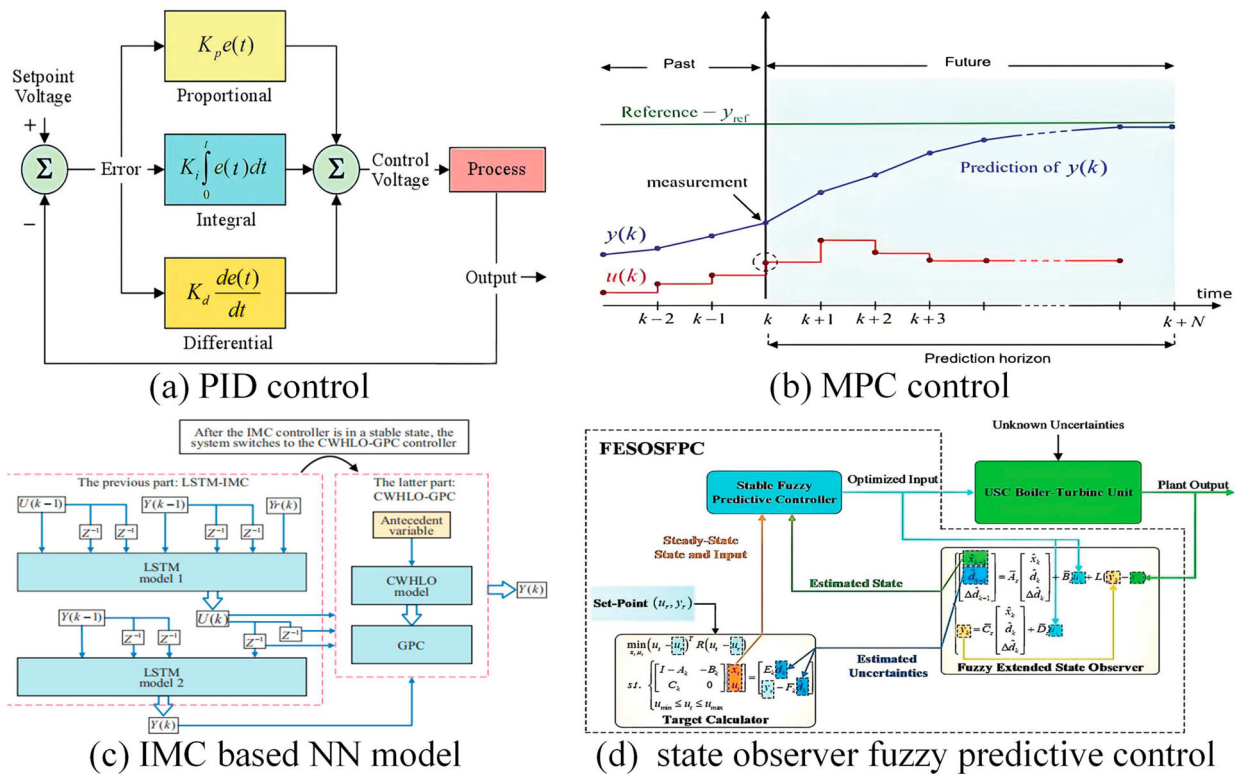


Figure 7. Control methodologies based on linear system theory.

The control strategies currently implemented in power plants remain predominantly fixed-parameter PID control systems (schematic diagram reference Figure 7(a)). In contrast, adaptive control systems are characterized by the ability to perform variable parameter tuning or self-tuning according to unit operating conditions, demonstrating superior load adaptability. Consequently, various self-tuning methods and optimization design techniques have emerged accordingly.

In response to the challenges of strong coupling and nonlinearity encountered by USC units during operation, an adaptive PID strategy is proposed based on model reference adaptive control (Kocaarslan & Cam, 2007b; Xu et al., 2018). By integrating the simplicity of conventional PID control with the agility of adaptive control, it dynamically optimizes controller parameters in real time through continuous comparison of actual output and the ideal model. This ensures rapid and stable system performance across varying load conditions. However, adaptive control algorithms require online estimation and tuning of multiple parameters (e.g. feedback/feedforward polynomials, compensator parameters), which demands expertise and may pose challenges for non-specialists. To enhance accessibility while maintaining robust performance, a novel fuzzy immune PID controller integrates PID control with intelligent optimization algorithms is proposed (Bouchebbat & Gherbi, 2017; Jagatheesan et al., 2017). It is used to enhance the control performance of steam temperature systems in thermal power plants. By simulating biological regulatory mechanisms, the controller establishes a nonlinear control architecture that dynamically strengthens control action for rapid response when system deviations are significant, while automatically suppressing overshoot as deviations diminish. Advancing the integration of artificial intelligence methodologies, a cutting-edge approach utilizing neural networks is proposed to enhance conventional PID control methods. This framework synergistically combines neural networks with evolutionary algorithms, leveraging the self-adaptive learning properties of neural networks and the stochastic global optimization capabilities of evolutionary algorithms to optimize PID controller parameters. Rigorous model validation and comparative simulation studies demonstrate the superior performance of this intelligent control approach, showcasing significant improvements in control precision, response time, and system stability (Si et al., 2024).

In recent years, robust control methods have been increasingly applied to the design of USC control systems to maintain superior performance even under significant uncertainties in the unit's mathematical model. Robust control excels in addressing system uncertainties while ensuring stability and performance, yet its design complexity, computational burden, and inherent conservatism may constrain its efficiency in certain practical applications (Ben-Abdenmour et al., 2002; Weng & Ray, 1997). Consequently, in real-world implementations, robust control strategies are often integrated with other control approaches to optimize performance and enhance system reliability and operational efficiency (Gu et al., 2022; Harutyunyan et al., 2023; Zheng et al., 2023).

One such approach is the application of robust H_∞ tracking control, which aims to ensure stable and optimal performance. This method has been widely researched for its ability to handle uncertainties while maintaining robust system operation (Akbari et al., 2019; J. Wu et al., 2010). To improve the operational flexibility of USC units, an enhanced internal-model robust adaptive control approach is proposed. By reducing the reliance on precise modelling, this method effectively addresses the control challenges in the dynamic conditions of USC units (L. Pan et al., 2020). For scenarios involving load-following operations and inevitable uncertainties, a nonlinear robust sliding mode controller is developed based on affine nonlinear control theory. This controller incorporates a precise gray-box multivariate coupled model of USC units, enabling effective coordinated control of once-through boiler systems (Z. Tian et al., 2018). Building upon the advanced nonlinear control framework for USC units, to addresses the rapid frequency regulation in modern power systems with high renewable penetration. A robust model predictive control strategy is proposed for condensate throttling in USC units, enabling fast reserve provision while maintaining operational constraints (X. Li et al., 2025).

Predictive control is a model-based optimization control algorithm that incorporates rolling implementation and feedback correction. As shown in Figure 7(b), its advantage lies in the adoption of a receding horizon optimization strategy, which enables real-time adjustments based on operational conditions while seamlessly integrating diverse optimization objectives into the control law (Qin & Badgwell, 1997; Schweidtmann et al., 2024; J. Wang & Ramirez-Mendoza, 2021).

The inherent strong nonlinearity of USC units makes conventional linear model-based predictive controller design fundamentally inadequate. As mentioned in the preceding modelling section of this paper, a multi-model approach is introduced where an array of local linear models collectively captures the system's nonlinear dynamics. This framework enables the design of localized predictive controllers for each linear sub-model, which are then intelligently coordinated through an adaptive scheduling strategy to form an integrated control scheme with optimized global performance (C. Cheng et al., 2020; X. Wu et al., 2013). As an effective tool for nonlinear dynamic modelling, neural networks (NNs) have been widely employed for USC modelling (Cui et al., 2020; Zhu et al., 2022). For controller design purposes, local linearization of NN models is commonly

adopted to obtain piecewise linear approximations, upon which linear predictive controllers can be developed while maintaining satisfactory control performance. Notably, internal model control (IMC) offers distinctive advantages in rapid regulation. Building upon accurate nonlinear NN models, researchers have developed IMC schemes that integrate with predictive controllers for USC systems (Lee et al., 2009). The schematic diagram can be found in Figure 7(c). Such hybrid approaches achieve both fast response and precise control, as demonstrated in recent studies (C. Cheng et al., 2024; Mohamed, 2024). The synergy between NN-based modelling and advanced control architectures has proven particularly effective for complex USC dynamics. Furthermore, predictive control frameworks have been augmented with additional techniques to improve system robustness. For instance, the incorporation of fuzzy extended state observers and mixed logical dynamic models (schematic diagram reference Figure 7(d)) has significantly enhanced disturbance rejection capabilities while achieving superior coordinated control performance (C. Chen et al., 2018; C. Yang et al., 2022). These methodological extensions effectively address critical challenges in USC unit operation by combining the predictive power of MPC with adaptive estimation and real-time adjustments to ensure optimal operational performance under varying conditions.

While the linear-based control strategies discussed above have been enhanced with multi-model and adaptive techniques, they ultimately face fundamental limitations when confronting the intrinsic nonlinearities of USC units. To achieve optimization across the full operational range, control methodologies that directly address and exploit these nonlinear characteristics are required. This motivates the exploration of control strategies grounded in nonlinear system theory.

4.2. Control methodologies based on nonlinear system theory

While there are numerous nonlinear control methodologies, the controlled plant of USC units exhibits intrinsic nonlinear characteristics. Theoretically, these nonlinear approaches possess the fundamental capability to address system-wide optimization across full operational ranges. However, given the inherent complexity of nonlinear methods, both their research outcomes and algorithmic practicality require more extensive investigation before full-scale implementation can be realized. Previously, based on predictive control, the design of global controller base on scheduling and real-time controller was introduced. Next, other methods will be introduced.

Feedback linearization represents a cornerstone methodology in nonlinear control theory. As shown in Figure 8(a) Through the application of state or output feedback, this approach enables the complete transformation of a nonlinear system into an exactly linearized counterpart, thereby fundamentally eliminating the intrinsic distinction between linear and nonlinear control problems (Gao et al., 1992; Yaniv, 1999).

Feedback linearization stands as a fundamental approach, is particularly effective for USC units where it facilitates pole-placement control design with wide operational range and improved dynamic performance (Daren & Zhiqiang, 2005). Extending beyond pure feedback strategies, the incorporation of feedforward compensation techniques has enhanced tracking performance in decentralized control architectures, particularly for multimachine systems where input-output linearization combines with feedforward action to improve dynamic response (De Tuglie et al., 2008). To address practical implementation challenges, robust control approaches employing approximate dynamic feedback linearization have been developed, utilizing extended state observers to estimate and compensate for system nonlinearities and disturbances while maintaining stability under parameter uncertainties (T. Yu et al., 2010). The most recent advances integrate observer-based integral control with multi-time-scale decomposition techniques, where innovative time-scale separation methods enable computationally efficient implementation while ensuring zero steady-state error through integral action and state estimation (Munje et al., 2018). Recent advances in neural-enhanced feedback linearization show particular promise for USC unit control. By integrating neural approximators with exact feedback linearization, these methods address the severe nonlinearities and wide operating ranges characteristic of USC boiler-turbine systems. The neural components adaptively compensate for USC-specific dynamics like steam parameter variations and material stress constraints, while maintaining the theoretical guarantees of linearization (Arif et al., 2013; Babaei et al., 2011; Fregene & Kennedy, 2000).

Fuzzy control has been widely implemented in control systems due to its superior capability in handling nonlinearity, time-varying dynamics, and system uncertainties. Its basic principle is shown in Figure 8(b) which simulates human experience reasoning, fuzzifies precise inputs, and performs logical reasoning based on a rule

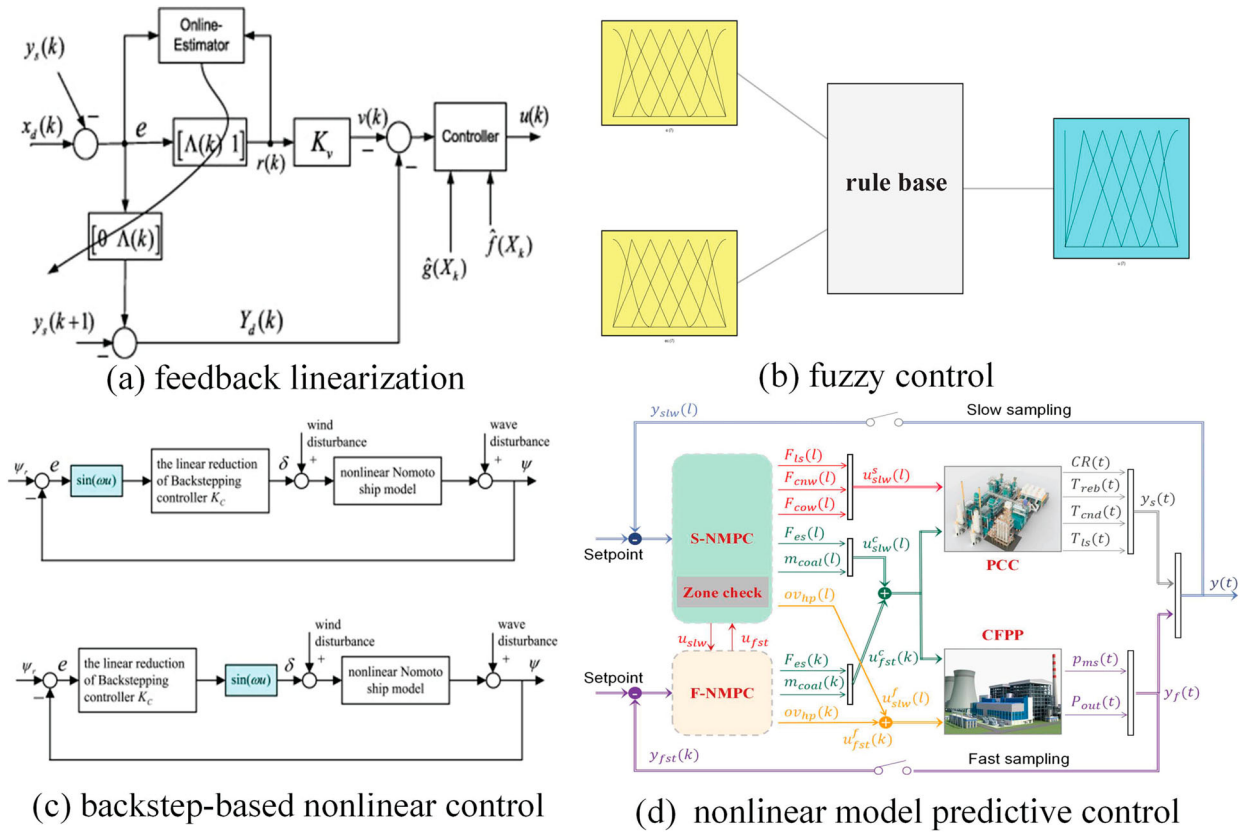


Figure 8. Control methodologies based on nonlinear system theory.

base. With the continuous advancement of control theory, fuzzy control methodologies have progressively integrated with other advanced control strategies to further enhance system robustness, adaptability, and economic efficiency (Garduno-Ramirez & Lee, 2000; Nguyen et al., 2019; H. H. Tang & Ahmad, 2024).

One prominent approach combines fuzzy logic with conventional PID control, where fuzzy gain scheduling dynamically adjusts PID parameters to adapt to varying operating conditions, as demonstrated in combustion control of coal-fired boilers (W. Li & Chang, 2000; Mohindru, 2024) and pH neutralization processes (Naseer & Khan, 2013). Another significant direction integrates fuzzy control with optimization-based methods, such as fuzzy model predictive control for handling system constraints and multi-objective optimization in power plants (X. Liu et al., 2022), and fuzzy-PSO algorithms for robust tuning of fractional-order processes (I. Pan et al., 2013; M. Tang et al., 2024). Furthermore, the fusion of fuzzy control with intelligent techniques has gained traction, including fuzzy-neural networks for adaptive rule adjustment in hybrid power systems (Natsheh & Albarbar, 2013), as well as fuzzy-reinforcement learning and fuzzy-deep learning architectures for combustion optimization, leveraging data-driven adaptation to improve dynamic response and operational precision (H. Liu et al., 2020; Zhan et al., 2021). These hybrid strategies collectively represent the cutting edge in control system design for thermal power plants, offering enhanced robustness, adaptability, and efficiency across diverse operational scenarios.

Contemporary research has increasingly focussed on sophisticated nonlinear control methodologies to address the inherent challenges of USC units. These approaches fundamentally preserve and exploit the system's nonlinear characteristics, eschewing conventional linearization approximations to achieve precise regulation.

Lyapunov-based stability control has emerged as a fundamental approach, where the strategic design of Lyapunov functions ensures system stability while effectively compensating for uncertainties and disturbances (De Queiroz, 2000; Dixon et al., 2002). Building upon this theoretical foundation, Bankole et al. developed an innovative Lyapunov stability-based scheduling methodology that has demonstrated exceptional performance in load dispatch optimization and carbon capture integration (Bankole et al., 2018; Heidarinejad

et al., 2012). The control challenges become more pronounced when addressing the complex nonlinearities inherent in boiler-turbine systems. This has led to the development of advanced techniques such as backstepping control, which offers a systematic solution for systems with intricate nonlinear dynamics. As shown in Figure 8(c), the work of Fang et al. exemplifies this approach through their development of a nonlinear adaptive control scheme that not only optimizes combustion efficiency but also significantly enhances load adaptability (Fang & Wei, 2011; Furtat & Tupichin, 2014, October). Recent advancements in control theory have further expanded the toolkit available to power plant operators, particularly through the evolution of nonlinear model predictive control. The implementation of these advanced strategies is exemplified by the development of dual-time-scale MPC frameworks (schematic diagram reference Figure 8(d)), which have shown remarkable success in optimizing post-combustion carbon capture systems (C. Yang et al., 2024; Y. Zhou, 2022; Zhuo, 2008).

These cutting-edge approaches represent a significant leap forward in control performance and operational efficiency, demonstrating how theoretical advancements in nonlinear control can be translated into practical solutions for modern power generation challenges.

Building upon the model-based linear and nonlinear control paradigms, the recent convergence of big data and computational intelligence has catalyzed the emergence of a new frontier: data-driven intelligent control. After systematically analyzing the control strategies of USC units based on both linear and nonlinear models, this paper further explores the application of artificial intelligence technologies in USC unit control. Unlike traditional model-dependent approaches, advanced intelligent control technology leverages data-driven methodologies and model self-adaptation to more effectively tackle the unique challenges of USC units.

4.3. Advanced intelligent control technology for USC units

After systematically analyzing the control strategies of USC units based on both linear and nonlinear models, this paper further explores the application of artificial intelligence technologies in USC unit control. Unlike traditional model-dependent approaches, AI-powered advanced intelligent control technology leverages data-driven methodologies and model self-adaptation to more effectively tackle the unique challenges of USC units, such as multivariate strong coupling and rapid wide-load dynamic conditions.

Faced with these challenges, reinforcement learning (RL) and deep learning (DL) have emerged as two pivotal, complementary paradigms driving intelligent control in USC units. Their functional roles, while distinct, are critically synergistic. In terms of core operational paradigm, RL is fundamentally dedicated to sequential decision-making and closed-loop optimization. Its agent, through continuous interaction with the environment, autonomously learns optimal dynamic adjustment strategies, essentially solving the problem of 'how to act'. This makes RL particularly well-suited for tackling multi-variable, strongly coupled control challenges under rapid load-changing conditions (Sprague et al., 2025). In contrast, DL is more focussed on feature perception and dynamic modelling. Leveraging the powerful nonlinear mapping capabilities of deep neural networks, it excavates high-dimensional features and complex nonlinear relationships from massive datasets to construct high-precision predictive models (USC Media Communications Lab, 2025). Regarding their implementation paradigms, DL model performance is heavily dependent on large volumes of high-quality, historically labelled data. Conversely, RL requires the collection of state-action sequence data through extensive trial-and-error exploration in high-fidelity simulation environments, which imposes significantly stricter demands on both model fidelity and learning algorithm data efficiency compared to DL (X. Wang & Huang, 2025).

Based on the above complementary characteristics, reinforcement learning has demonstrated remarkable advantages in the control of coal-fired power generation units, particularly in autonomous decision-making and optimized control under complex operating conditions. Wang et al. proposed a deep reinforcement learning framework that was successfully applied to wide-load operation optimization in USC units, achieving a 12% improvement in thermal efficiency across the 30%-100% load range (Z. Wang et al., 2024). The DeepThermal project developed an offline RL framework, which learns optimal control strategies through model-environment interactions and has been implemented in four major coal-fired power plants in China for combustion efficiency optimization (Zhan et al., 2021). Notably, to address the control challenges of denitrification systems under variable load conditions, a transfer reinforcement learning approach is introduced, which effectively resolves control strategy generalization across different operating conditions through knowledge

Table 5. Control review summary (coverage 1978–2025).

Control methodologies	Advantages	Disadvantages
Control methods based on linear system theory (Kocaarslan & Cam, 2007b–H. Yang et al., 2025)	Currently stands as the most technologically mature and extensively solution across industrial applications	Exhibits fundamental limitations in adapting to transient operational regimes, rendering linear controllers particularly susceptible to excitation-induced instability
Control methods based on nonlinear system theory (Cao et al., 1992–Zhou, 2022)	Fundamentally resolves the systemic challenges of large-scale, full-operational-range performance optimization	Requiring precise mathematical models, facing challenges in high-noise scenarios and theoretical complexity
Advanced intelligent control technology (Z. Wang et al., 2024–Bisset et al., 2023)	A model-independent control paradigm exhibiting inherent robustness against parametric uncertainties and external disturbances	The applicability remains inferior to linearization-based methods; the majority of current research confined to simulation-based validation

transfer mechanisms (Fu et al., 2020; X. Sun et al., 2024). These findings demonstrate that RL can effectively overcome the limitations of traditional control methods in managing nonlinear, time-varying systems.

Deep learning technology, with its powerful feature extraction and pattern recognition capabilities, provides innovative solutions for coal-fired unit control. In neural network-based control, Suresh et al. combined artificial neural networks (ANNs) with genetic algorithms to optimize operational parameters in high-ash USC units (Suresh et al., 2011). A recent study proposed a PSO-LSTM hybrid model, employing LSTM network to process time-series data, significantly enhancing automatic generation control performance (Hao et al., 2024). Stružnik et al. and Yao et al. developed AI-based combustion control schemes for power units. Both studies featured real-time prediction, optimization, and closed-loop control. Stružnik et al. used LNFN and MLP to predict NO_x and adjust urea injection, achieving emission compliance while reducing urea and water consumption. Yao et al. employed a feedforward neural network and genetic algorithm to optimize combustion parameters, resulting in reduced NO_x emissions, ammonia consumption, and improved heat rate (Stružnik et al., 2021; Yao et al., 2023).

While RL and deep learning exhibit transformative potential for USC unit automation, critical hurdles around data efficiency, computational overhead, and operational safety must be resolved. Strategic investments in hybrid modelling paradigms, edge AI hardware, and explainability engineering will be pivotal for transitioning these technologies from lab prototypes to industrial-grade solutions. Looking ahead, artificial intelligence is poised to revolutionize optimal control in thermal power generation, with breakthroughs expected in wide-load operation optimization, combustion efficiency enhancement, and intelligent coordinated control systems (Bisset et al., 2023).

Table 5 summarizes the survey coverage of different types of control methods from 1978 to 2025. From a practical perspective, the control strategies for USC units can be broadly evaluated in terms of stability, adaptability, and optimization capability.

Linear control methods remain the most mature and widely applied in industrial practice. They offer simple implementation, reliable stability, and satisfactory steady-state accuracy under near-linear operating conditions. However, their capability becomes limited when addressing the strong nonlinearity and multivariable coupling inherent to USC systems. Nonlinear control methods demonstrate better adaptability to complex dynamics and parameter variations, with superior disturbance rejection and transient performance. The main challenges lie in their high modelling complexity and demanding real-time computational requirements. Intelligent control methods exhibit remarkable self-learning and adaptive optimization abilities, capable of achieving near-optimal performance under wide-load, time-varying conditions. Simulation and pilot-scale studies indicate significant advantages in improving thermal efficiency and emission control. Nevertheless, their industrial deployment is still limited due to data reliability, safety, and computational constraints.

In summary, linear control offers the most stable performance for existing engineering applications, nonlinear control provides superior dynamic regulation under complex and variable conditions, and intelligent control holds the greatest potential for achieving high-dimensional optimization and adaptive operation in next-generation USC unit control systems.

5. Proposed new trends

USC power generation technology, as the cornerstone of clean coal-fired power systems, is currently standing at the forefront of an unprecedented intelligent transformation. Through systematic review of existing

research, we have not only witnessed groundbreaking advancements in intelligent control technologies but also identified a spectrum of highly promising future research directions that may potentially redefine the control paradigm for next-generation clean coal power systems.

- (1) Extensive research findings have consistently highlighted the significant yet largely unexplored potential of advanced system identification methodologies for application in USC units. Current academic investigations reveal that several sophisticated model architectures - including but not limited to Wiener models, Hammerstein-Wiener nonlinear structures, enhanced Wiener formulations, and Hammerstein paradigms – merit particularly rigorous examination and experimental validation (Hou et al., 2024; F. Li et al., 2024; Mohamed et al., 2020). This recommendation stems from their empirically demonstrated capacity to deliver substantial performance enhancements when applied to modelling challenges in comparable energy conversion systems. The direct implementation of these identification techniques for comprehensive USC unit characterization presents a valuable research opportunity, while simultaneously creating a robust framework for systematic comparative assessment against conventional and emerging alternative modelling approaches.
- (2) A notable ‘imbalance phenomenon’ prevails in current intelligent control research: studies based solely on neural networks account for over 75% of publications, while those integrating physical models constitute less than 15%. This disparity is catalyzing a transformative research opportunity – the development of ‘symbiotic systems’ that harmoniously combine physical models with neural networks. The inherent stability and interpretability of physical models under emergency conditions exhibit natural complementary advantages with neural networks’ precision in tracking large load variations, creating a synergistic relationship akin to quantum entanglement. We anticipate that the next breakthrough in controller design will emerge from ‘Physics-Informed Neural Hybrid Controllers’. This innovative architecture simultaneously leverages the observable intermediate variables of physical models (such as locally measured steam-water system parameters) while capitalizing on neural networks’ exceptional capability to capture nonlinear dynamics (Alhajeri et al., 2022; Antonelo et al., 2024; Roehrl et al., 2020).
- (3) The optimization of startup processes represents an underexplored yet highly valuable research frontier in USC power generation. Contrary to conventional understanding, approximately 15% of potential efficiency improvements remain untapped during plant startup – equivalent to annual savings of nearly 2000 tons of standard coal for a typical 1000 MW unit. Multi-model predictive control (MMPC) demonstrates particular promise in this domain, where the construction of a comprehensive linear state-space model cluster encompassing all startup conditions, combined with adaptive weight allocation algorithms, enables intelligent optimization of startup curves. We specifically propose an innovative ‘digital twin + MMPC’ integrated research approach, wherein thousands of startup scenarios are simulated in a virtual power plant environment to train optimal control strategies.

6. Conclusion

This paper presents a comprehensive and systematic review of the state-of-the-art modelling, simulation, and intelligent control technologies for USC units. Through in-depth analysis and classification of existing literature, the following key conclusions are drawn:

Modelling methodologies:

- *First-principles modelling* remains essential for characterizing USC unit thermodynamics, particularly under extreme operating conditions. Based on fundamental conservation laws, these models provide crucial insights into boiler-turbine dynamics but face three key limitations: (1) oversimplified phase transition representations in once-through boilers; (2) inadequate handling of multivariable coupling effects; and (3) computational challenges for real-time implementation. While distributed parameter and CFD-coupled approaches have improved accuracy, gaps persist in modelling full-range dynamics, especially during rapid load transitions.
- *Data-driven and hybrid modelling approaches* have transformed USC modelling through neural networks’ nonlinear learning capability and fuzzy systems’ uncertainty handling. Physics-informed neural networks represent a significant advance by embedding conservation laws into network architectures, combining

physical interpretability with data-driven adaptability. Current research focuses on self-adaptive models for equipment aging and multi-scale frameworks integrating component and plant-level dynamics, offering promising directions for real-time monitoring and predictive maintenance.

Control strategies:

- *Conventional PID and linear control methods* are increasingly inadequate for addressing the complex non-linear dynamics and stringent operational requirements of USC units. These methods, while widely used in the past, struggle to cope with the high variability and unpredictability inherent in modern power systems.
- *Advanced control paradigms*, such as model predictive control, robust H_∞ control, and backstepping-based nonlinear control, have achieved significant performance improvements in load-following capability, emission reduction, and operational flexibility. These approaches provide a more sophisticated and adaptive way to manage the complex interactions within USC units, leading to enhanced overall system performance.
- *Intelligent control technologies*, including reinforcement learning, deep learning, and fuzzy logic-based systems, have emerged as transformative solutions for real-time optimization and autonomous decision-making, particularly under variable renewable energy integration scenarios. These technologies leverage data-driven insights and machine learning algorithms to optimize control strategies dynamically, offering relatively ideal efficiency and responsiveness.

Future research directions:

- The recommendations for future research focus on the novel directions proposed in Section 5, highlighting cutting-edge intelligent technologies for parameter identification and control applications that have yet to be implemented in USC units. It is believed that these concise yet forward-looking suggestions will pave the way for researchers to conduct more in-depth investigations in this field, provided they possess well-equipped research facilities and adhere to rigorous academic standards ultimately yielding enhanced and transformative outcomes.

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Data availability statement

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Author contributions

CRedit: **C. L. Cheng**: Conceptualization, Investigation, Methodology, Writing – original draft; **C. Peng**: Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Writing – review & editing; **F. Chu**: Data curation, Formal analysis, Validation; **Y. J. Fang**: Data curation, Resources

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